Day 044 | Unit 03 | Teacher plan

Integers and Rational Numbers

Solve Coordinate-Plane Problems

Standards: 6.NS.C.6b, 6.NS.C.6c, 6.NS.C.8

Learning target: I can solve Coordinate-Plane Problems, and explain and check my reasoning.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Start at the origin. Move horizontally by x, then vertically by y. Quadrant signs are I (+,+), II (-,+), III (-,-), IV (+,-). A point on an axis belongs to no quadrant. Across the x-axis, keep x and reverse the sign of y. Across the y-axis, reverse x and keep y. Across both axes, reverse both signs. The reflected point stays the same distance from the axis. Plot both points. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units. On opposite sides of zero, add distances to zero. On the same side, subtract magnitudes.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 1, 7

Day 044 | Unit 03 | Lesson

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01 / Learn

Start at the origin. Move horizontally by x, then vertically by y. Quadrant signs are I (+,+), II (-,+), III (-,-), IV (+,-). A point on an axis belongs to no quadrant. Across the x-axis, keep x and reverse the sign of y. Across the y-axis, reverse x and keep y. Across both axes, reverse both signs. The reflected point stays the same distance from the axis. Plot both points. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units. On opposite sides of zero, add distances to zero. On the same side, subtract magnitudes.

Worked example 1

Plot P(7, -2) on a coordinate plane and name its quadrant.

1. Start at the origin.

2. Move horizontally by x, then vertically by y.

3. Move 7 right, then 2 down.

4. State the result in the form requested: Quadrant IV.

Answer: Quadrant IV

Worked example 2

Plot A(-4, -3), B(5, -3), and C(-4, 3). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.

1. Plot both points.

2. AB is horizontal: 4 + 5 = 9.

3. AC is vertical: 3 + 3 = 6.

4. Each segment crosses an axis.

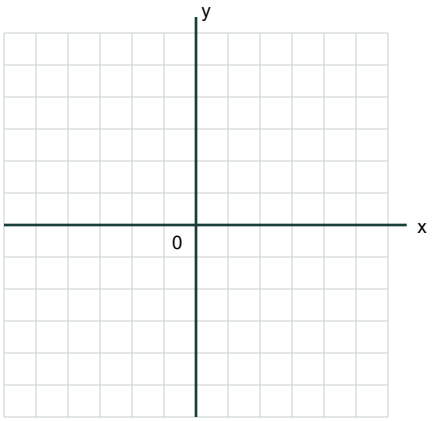
Answer: AB = 9 units; AC = 6 units

Day 044 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

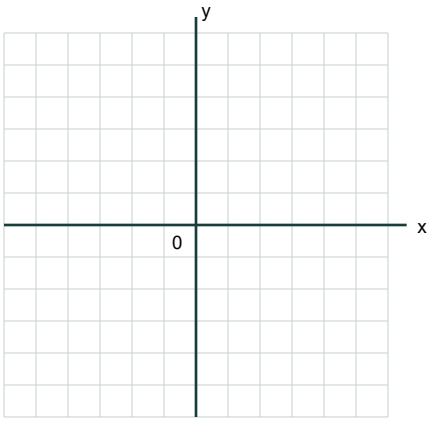
Plot P(-6, -3) on a coordinate plane and name its quadrant.



Label the scale and units.

02 / Constructed response

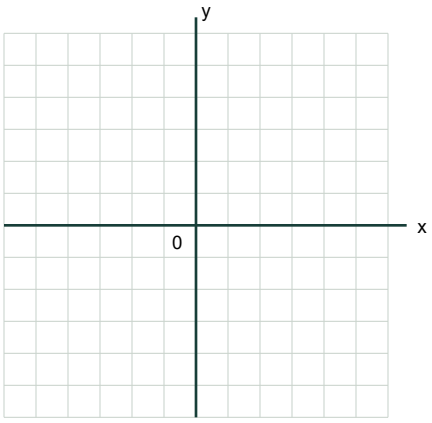
Plot (-5, -3), reflect it across y-axis, and give the reflected coordinates.



Label the scale and units.

03 / Constructed response

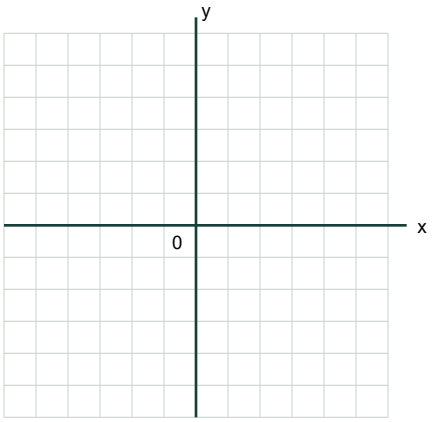
Plot (5, -3) and (5, -8). Find their distance.



Label the scale and units.

04 / Constructed response

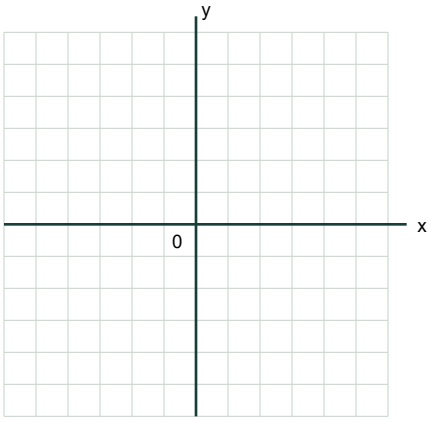
Plot P(8, -5) on a coordinate plane and name its quadrant.



Label the scale and units.

05 / Constructed response

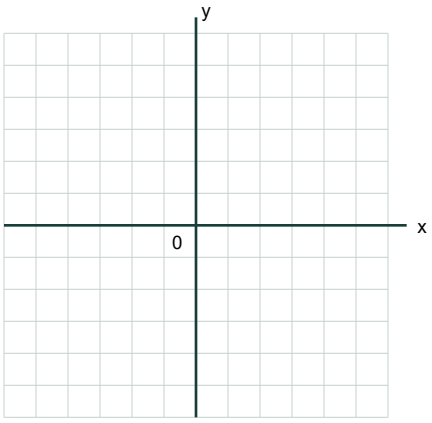
Plot (-4, 2), reflect it across x-axis, and give the reflected coordinates.



Label the scale and units.

06 / Constructed response

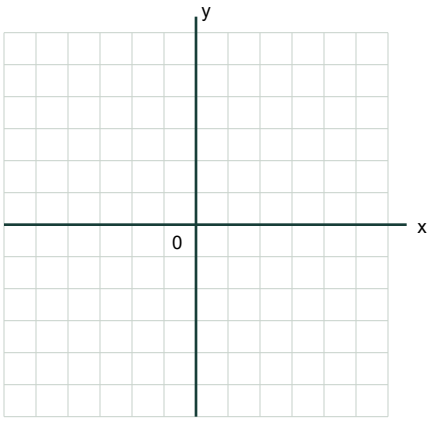
Plot (-8, 3) and (2, 3). Find their distance.



Label the scale and units.

07 / Constructed response

Plot P(-5/2, -2) on a coordinate plane and name its quadrant. Then plot Q on the y-axis at the same height and name its coordinates.



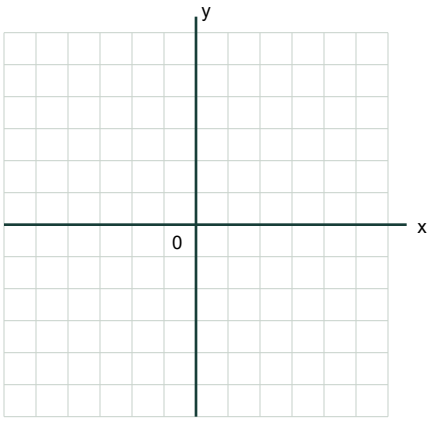
Label the scale and units.

08 / Multiple select

Point P is (-5, 4). Select all correct reflections.
A. Across the y-axis: (5, 4).
B. Across the x-axis: (-5, -4).
C. Across the x-axis: (5, 4).
D. Across both axes: (-5, 4).

09 / Coordinate plotting and numeric entry

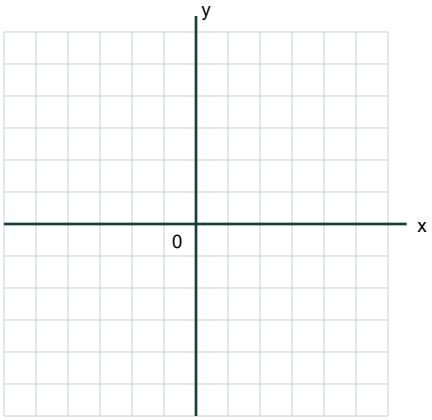
Plot A(-6, -3), B(5, -3), and C(-6, 3). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

10 / Constructed response

Plot P(3, -7/2) on a coordinate plane and name its quadrant. Then plot Q on the y-axis at the same height and name its coordinates.



Label the scale and units.

Day 044 | Unit 03 | Answer key

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1. Quadrant III

Move 6 left, then 3 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

2. (5, -3)

Reverse the x sign: (5, -3).

6.NS.C.6b | Constructed response

3. 5 units

Only one coordinate changes. Same side of 0: subtract magnitudes to get 5.

6.NS.C.8 | Constructed response

4. Quadrant IV

Move 8 right, then 5 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

5. (-4, -2)

Reverse the y sign: (-4, -2).

6.NS.C.6b | Constructed response

6. 10 units

Only one coordinate changes. Opposite sides of 0: 8 + 2 = 10.

6.NS.C.8 | Constructed response

7. Quadrant III; Q(0, -2)

Move 5/2 left, then 2 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

8. A, B - Across the y-axis: (5, 4).; Across the x-axis: (-5, -4).

The x-axis reflection changes y only; the y-axis reflection changes x only. Across both gives (5, -4).

6.NS.C.6b | Multiple select

9. AB = 11 units; AC = 6 units

AB is horizontal: 6 + 5 = 11. AC is vertical: 3 + 3 = 6. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

10. Quadrant IV; Q(0, -7/2)

Move 3 right, then 7/2 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response